Hyperspectral Remote Sensing

IRS Chapter 14.1-14.10

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Sections

1. **Foundations** (14.1-14.4): spectroscopy, calibration, and AVIRIS
2. **Image cubes and abundance mapping** (14.5-14.7): libraries, noise reduction, and endmembers
3. **Advanced analysis** (14.8-14.10): mixtures, spectral matching, and interpretation

Section 1 | Chapter 14.1-14.4

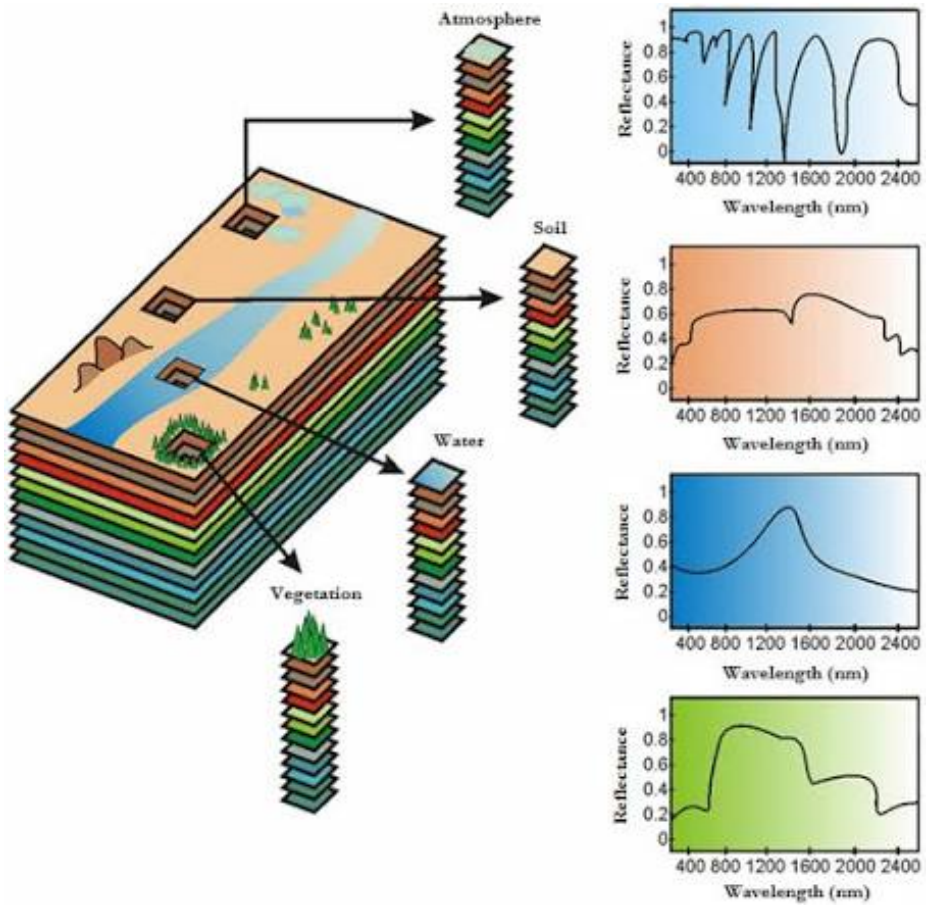
Foundations

Spectroscopy, calibration, and hyperspectral sensors

Introduction to Hyperspectral Remote Sensing

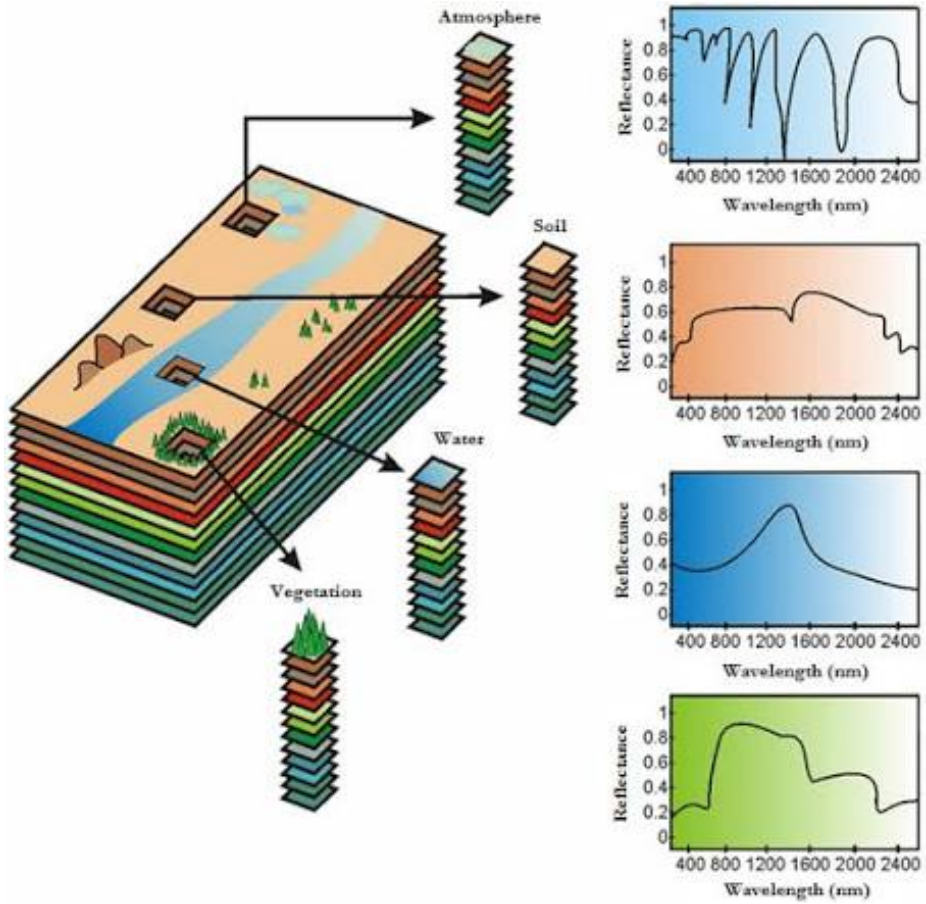
Narrow bands connect remote sensing to spectroscopy. We first examine calibration and AVIRIS, then move to image cubes, libraries, and unmixing.

- **Hyperspectral** vs. conventional remote sensing
- Uses **many narrow spectral channels** (often 200+)
- Channels as narrow as **10 nm**
- Enables **fine spectral resolution**



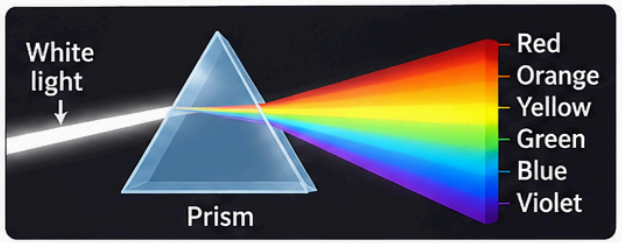
Spectroscopy and Hyperspectral Data

- **Spectroscopy** enhances material identification
- Uses **detailed spectral data** for analysis
- Enables **matching spectra to libraries**
- Supports atmospheric correction using information within the image



Isaac Newton (1666)

Separated white light into a spectrum with prisms.

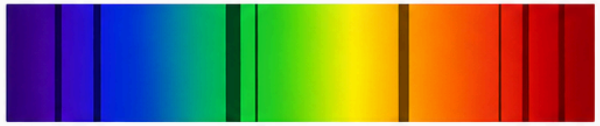


Impact on RS:

Established that light carries wavelength-specific information, the basis of all spectral sensing.

William Wollaston (1802)

Observed dark lines (gaps) in the solar spectrum.



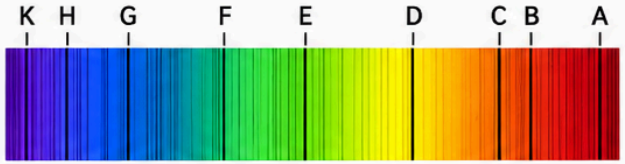
Solar spectrum (Wollaston's observation)

Impact on RS:

Revealed that spectra contain diagnostic absorption features that can be used to identify materials.

Joseph Fraunhofer (1814–1817)

Systematically measured and identified hundreds of solar absorption lines (Fraunhofer lines).



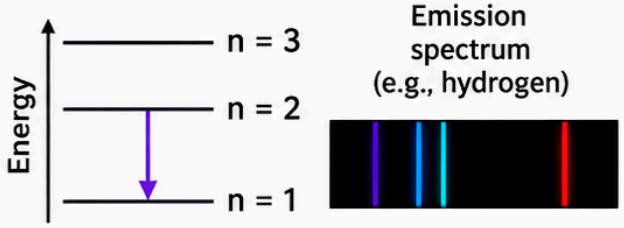
Fraunhofer lines in the solar spectrum

Impact on RS:

Provided a precise, repeatable spectral reference, enabling accurate wavelength calibration for remote sensing instruments.

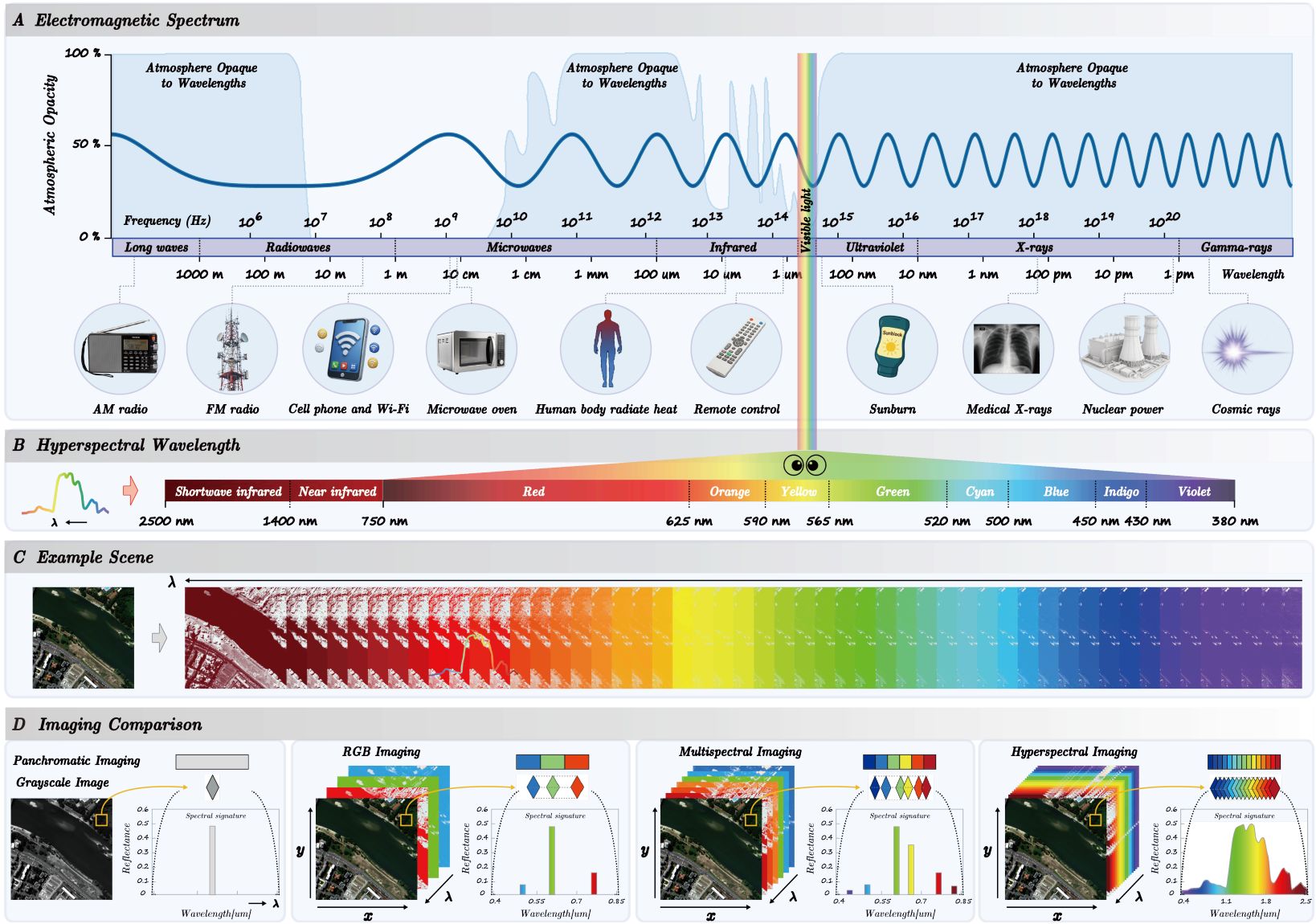
Niels Bohr (1913)

Linked spectral lines to quantized electron energy levels in atoms.

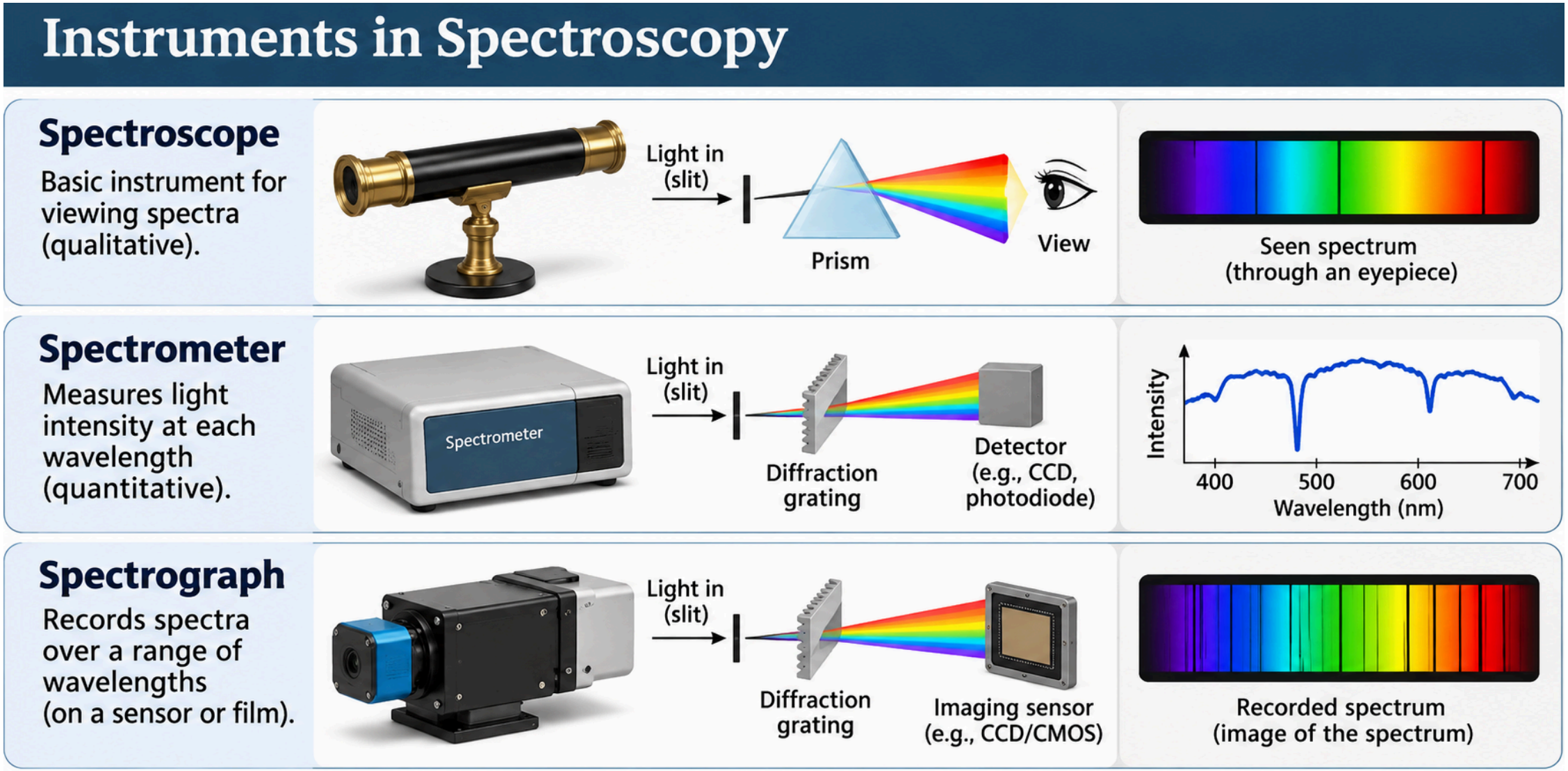


Impact on RS:

Explained the physical origin of spectral lines, enabling identification of elements and molecules (e.g., minerals, gases) from their spectral signatures.

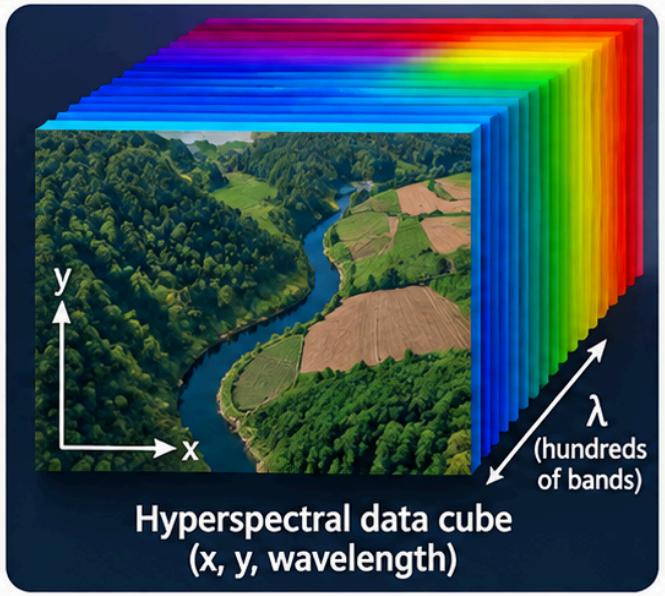

A Electromagnetic Spectrum
100 %
50 %
0 %
Atmospheric Opacity
Atmosphere Opaque to Wavelengths
Atmosphere Opaque to Wavelengths
Atmosphere Opaque to Wavelengths
Visible light
Frequency (Hz)
10^6 10^7 10^8 10^9 10^10 10^11 10^12 10^13 10^14 10^15 10^16 10^17 10^18 10^19 10^20
Long waves
Radiowaves
Microwaves
Infrared
Ultraviolet
X-rays
Gamma-rays
1000 m 100 m 10 m 1 m 10 cm 1 cm 1 mm 100 um 10 um 1 um 100 nm 10 nm 1 nm 100 pm 10 pm 1 pm
Wavelength
AM radio
FM radio
Cell phone and Wi-Fi
Microwave oven
Human body radiate heat
Remote control
Sunburn
Medical X-rays
Nuclear power
Cosmic rays
B Hyperspectral Wavelength
λ
Shortwave infrared
Near infrared
Red
Orange
Yellow
Green
Cyan
Blue
Indigo
Violet
2500 nm
1400 nm
750 nm
625 nm
590 nm
565 nm
520 nm
500 nm
450 nm
430 nm
380 nm
C Example Scene
λ
D Imaging Comparison
Panchromatic Imaging
Grayscale Image
RGB Imaging
Multispectral Imaging
Hyperspectral Imaging
Spectral signature
Reflectance
Wavelength[um]
x
y
λ

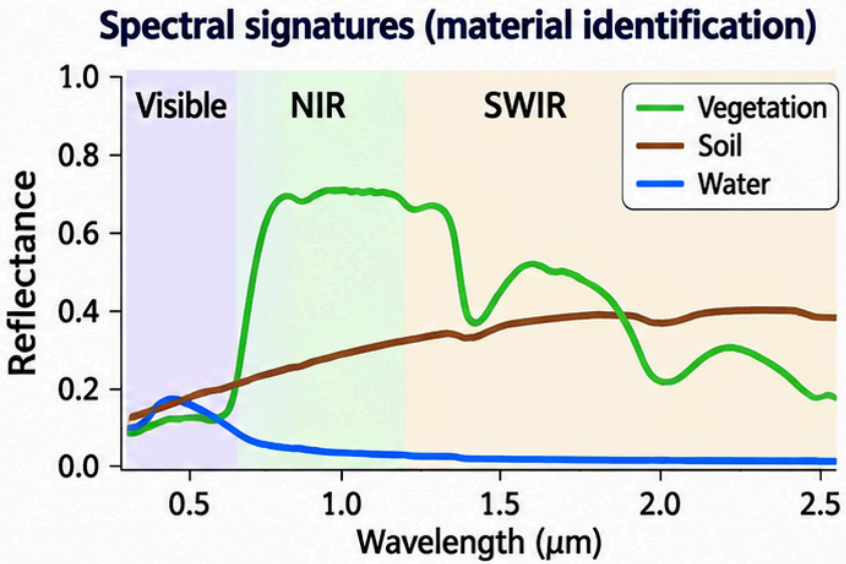
Instruments in Spectroscopy



Hyperspectral Imaging Applications

- Focuses on **identifying materials** through spectral signatures.
- Extends beyond visible wavelengths to **NIR and SWIR** (≈ 0.7–2.5 μm).
- Used for **Earth surface analysis** (e.g., vegetation, soil, water, minerals) and **atmospheric correction**.





Earth Surface Analysis

Vegetation (health, species) | Minerals (composition) | Water (quality, mapping)

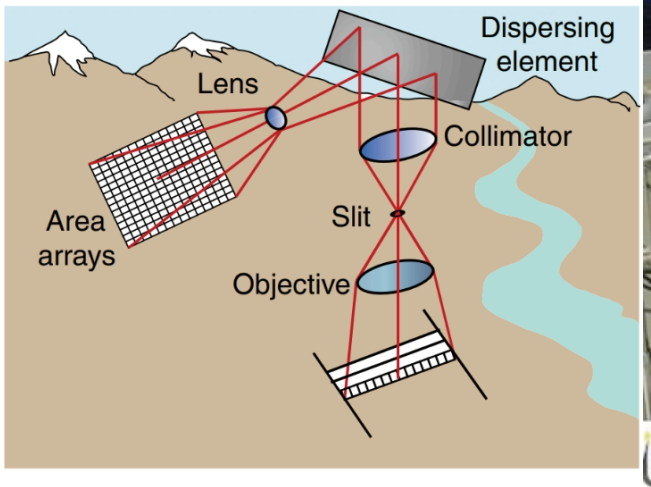
Atmospheric Correction

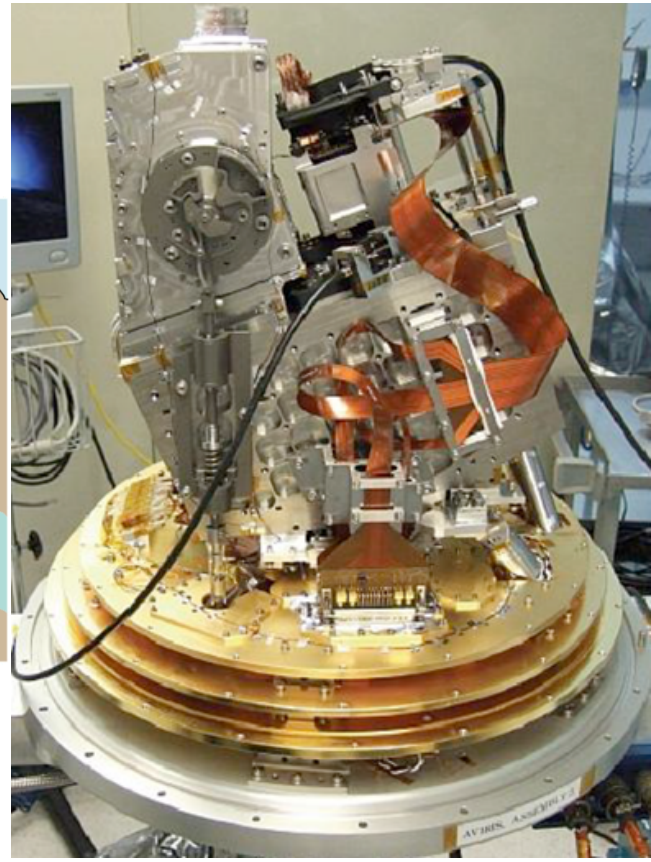
Importance of Calibration

- Calibration **ensures data accuracy** across channels
- Essential for **consistent spectral measurements**
- Addresses **instrument and environmental variability**

AVIRIS: A Key Hyperspectral Instrument

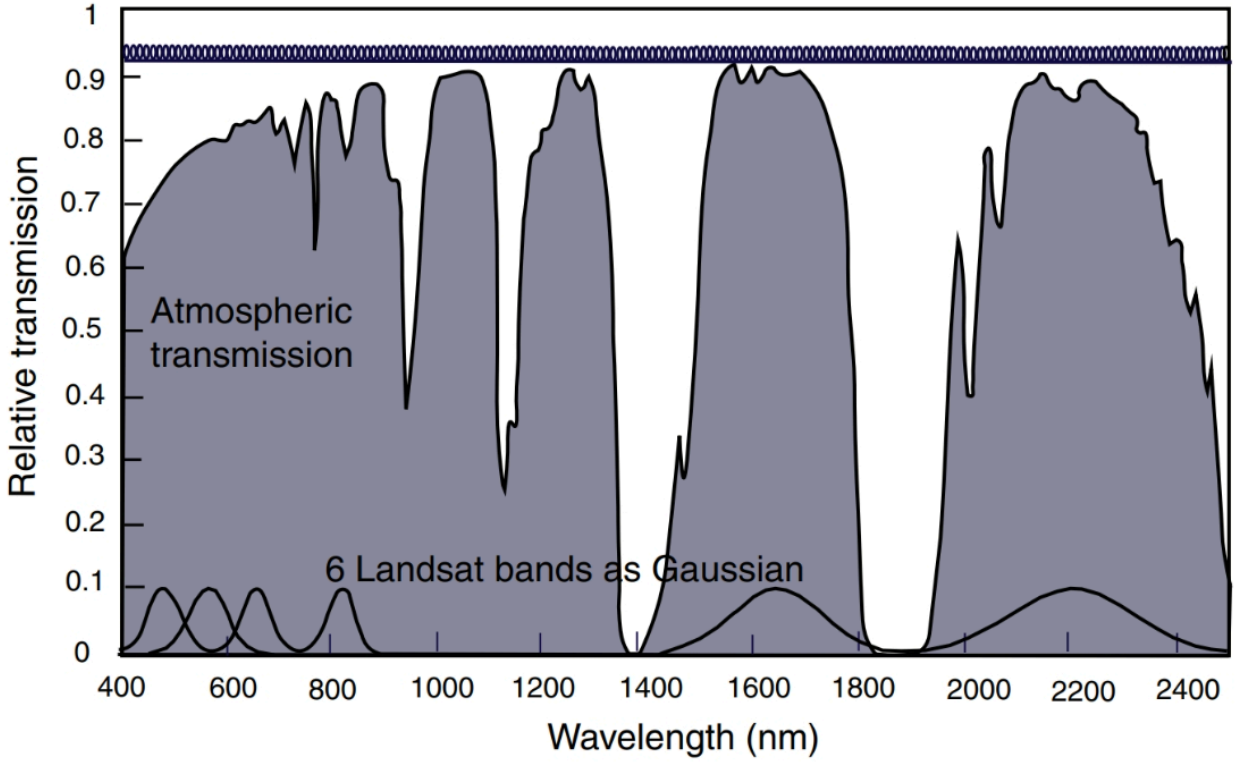
- Developed by **NASA/JPL** in the 1980s
- Collects **224 channels** across 400-2500 nm
- Enables **high-resolution hyperspectral imaging**





AVIRIS Spectral and Spatial Range

- Spectral range: **400-2500 nm**
- Channel width: approximately **10 nm**
- Spatial resolution varies by altitude:
  - High altitude: approximately **20 m** per pixel



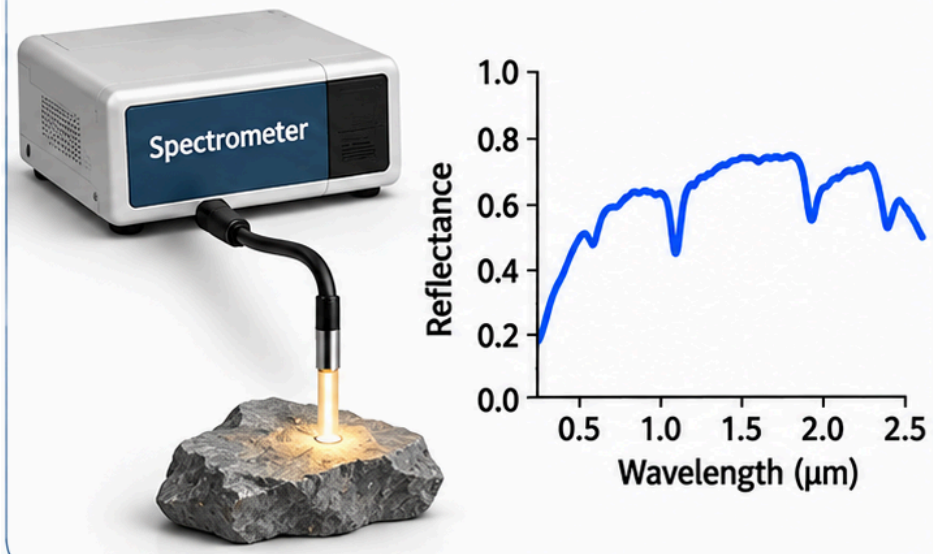
Hyperspectral Data and Material Analysis

- Enables **precise material matching** to spectral libraries
- Common in **mineralogy, agriculture, and environmental studies**
- Can support correction for **atmospheric effects**

Classical Spectroscopy vs. Hyperspectral Imaging

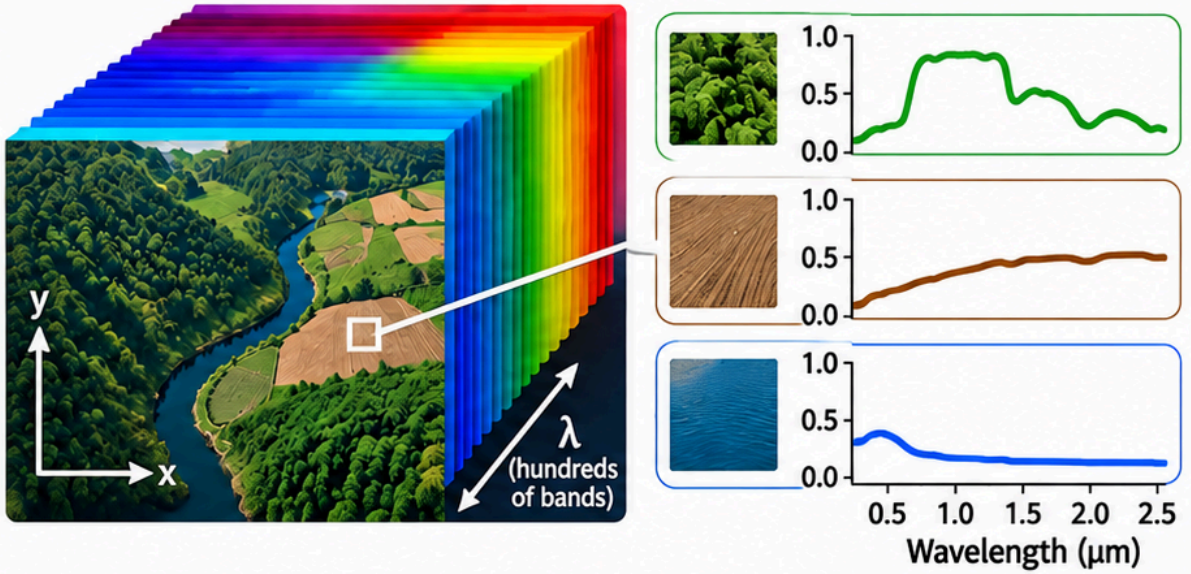
Classical Spectroscopy

Analyzes single-point targets (one spectrum at a time).



Hyperspectral Imaging

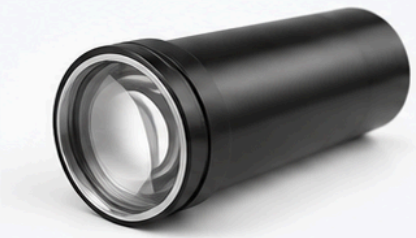
Captures a spectrum at every pixel over an extensive area.



→ Applied to **large surface areas** rather than isolated points.

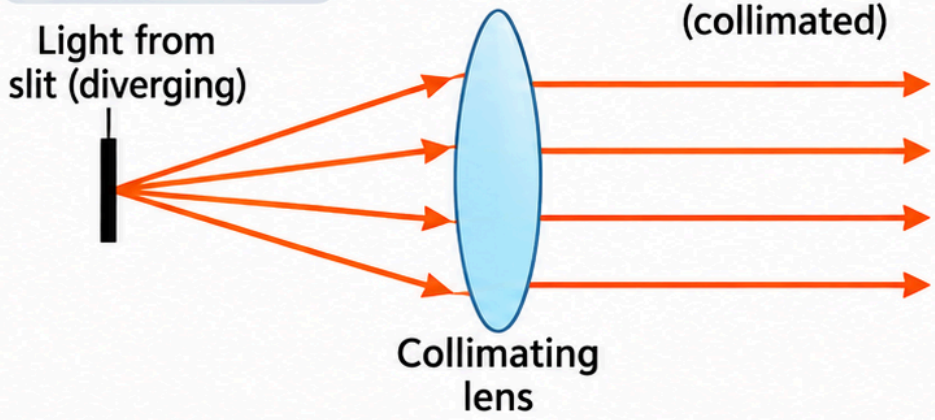
AI-generated illustration

Collimator



Creates a parallel ("collimated") beam from a diverging beam.

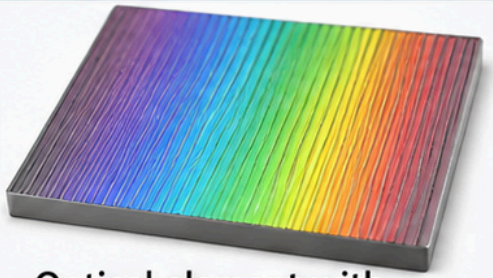
How it works



Why it matters

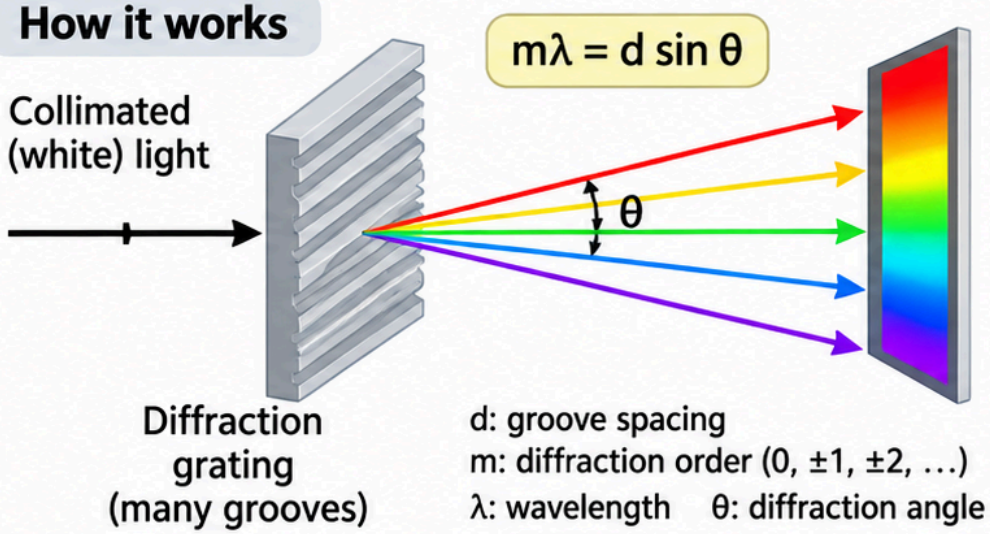
- Provides a uniform, parallel beam for the diffraction grating.
- Improves spectral resolution and measurement accuracy.
- Typically a lens or mirror (e.g., collimating lens or off-axis mirror).

Diffraction Grating



Optical element with many closely spaced grooves that separates light by wavelength via diffraction.

How it works

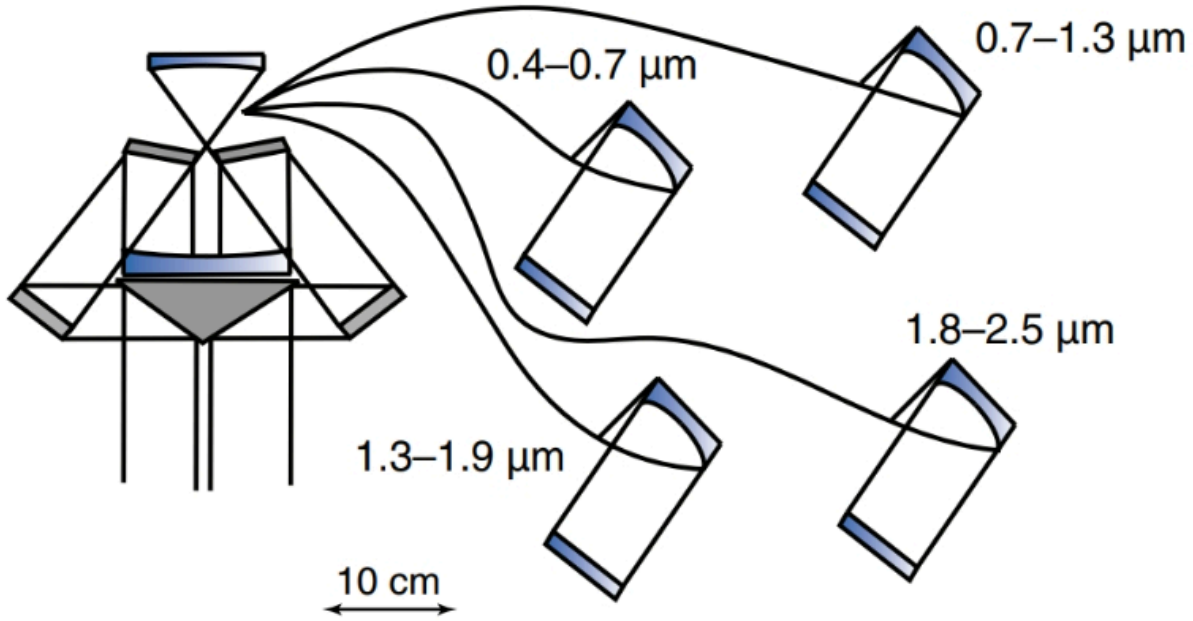


Key points

- A periodic surface (grooves) causes constructive interference at specific angles.
- Different wavelengths leave at different angles (dispersion).
- The grating equation: $m\lambda = d \sin \theta$.

Components of Hyperspectral Sensors

- **Diffraction grating**: Separates light into spectral bands
- **Detector arrays**: Capture visible, NIR, and SWIR ranges
- Fine **spectral, spatial, and radiometric resolution**



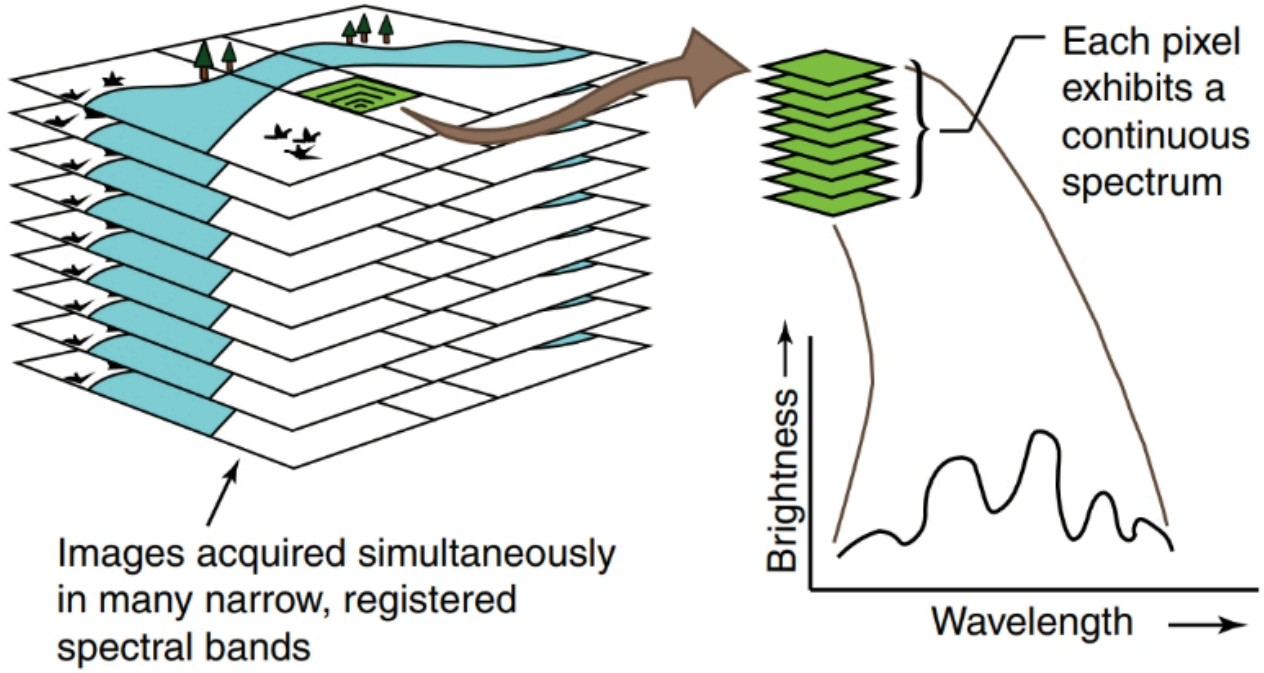
Section 2 | Chapter 14.5-14.7

Image Cubes and Abundance Mapping

Spectral libraries, noise reduction, and endmembers

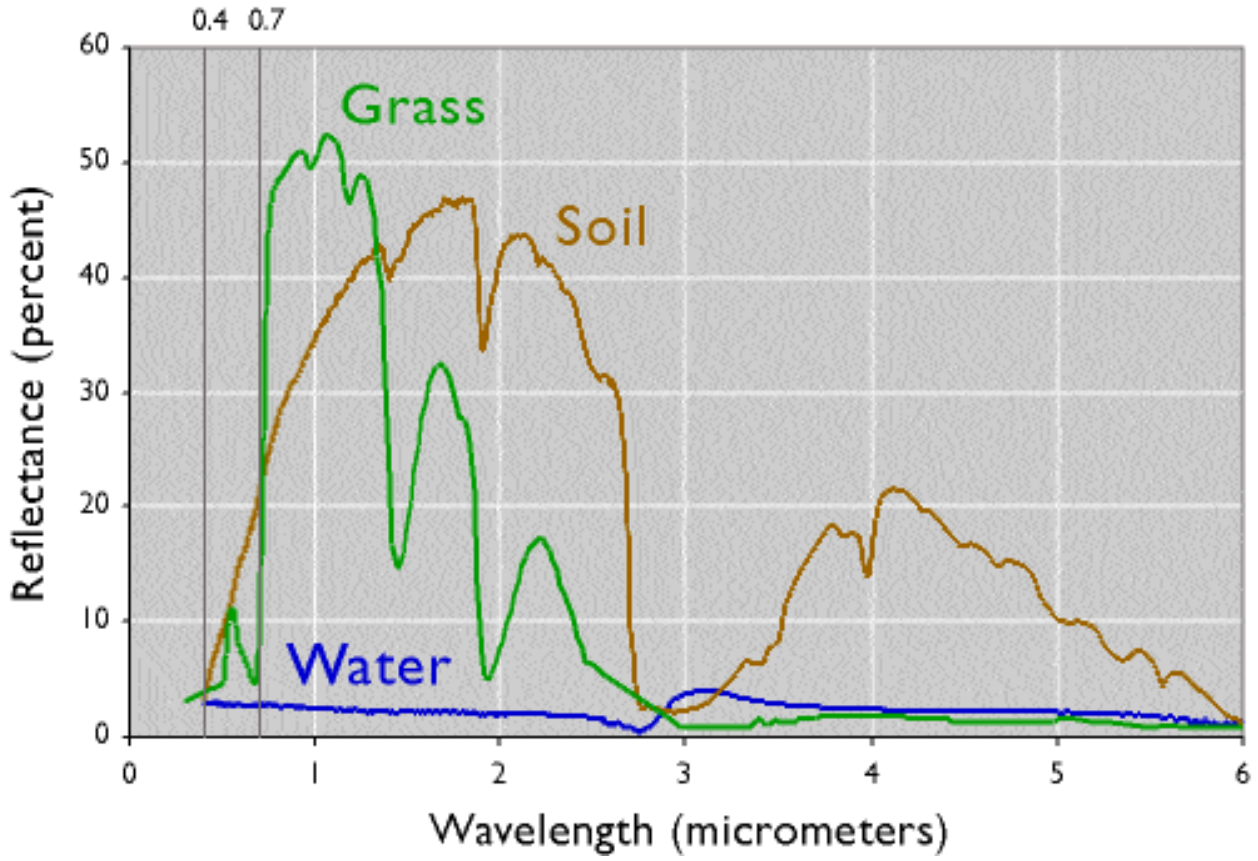
The Hyperspectral Image Cube

- **Image cube**: 3D structure in hyperspectral data
- **X, Y**: Spatial dimensions (map)
- **Z**: Stacked spectral bands (wavelengths)
- Captures **spectral profile for each pixel**



Understanding Spectral Libraries

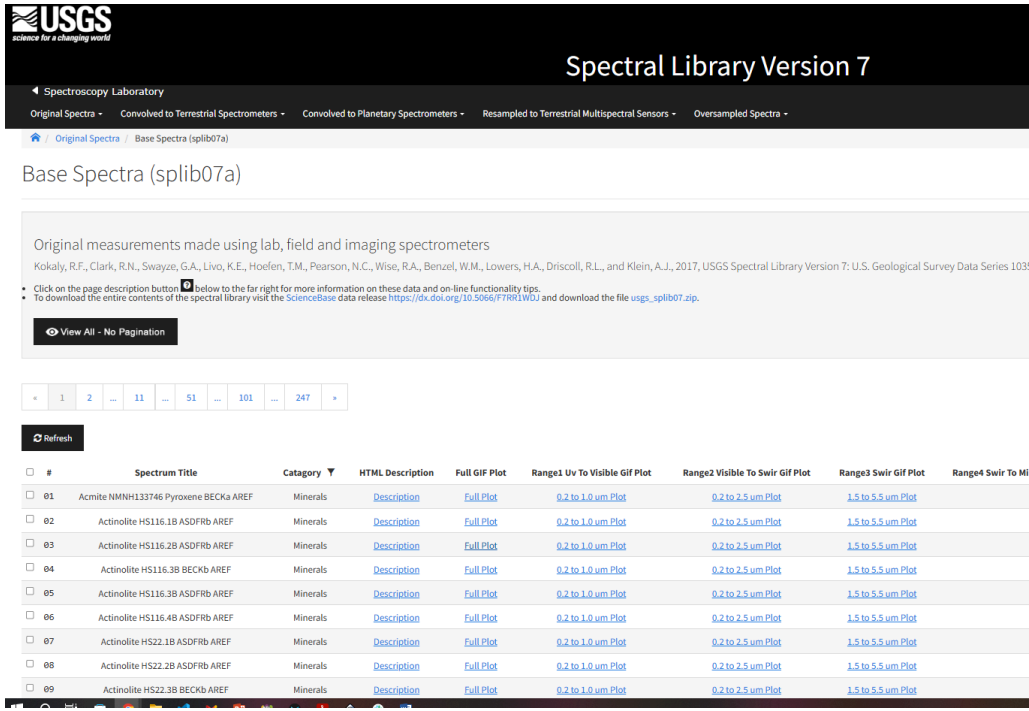
- **Spectral libraries**: Reference databases for spectral analysis
- Store spectra from **lab and field conditions**
- Data from **minerals, vegetation, materials**, etc.
- Essential for **material identification**



Data Sources for Spectral Libraries

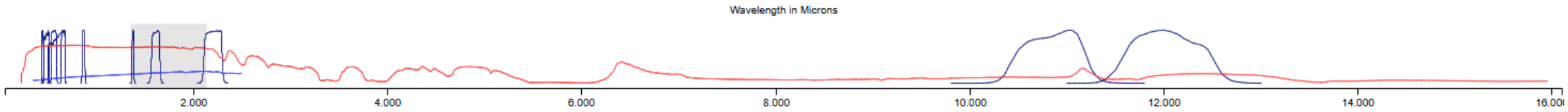
- Data from **laboratory** and **field measurements**
- Includes **minerals, vegetation, materials**, etc.
- Acquired under **controlled and natural conditions**
- Represent **varied terrain and climates**

Libraries compile spectra from diverse sources to support hyperspectral analysis.



Standardization in Spectral Libraries

- **Standardization** enables consistent use across applications
- Spectra must be comparable despite **instrument and condition differences**
- Records include **instrument details, conditions, and surface characteristics**

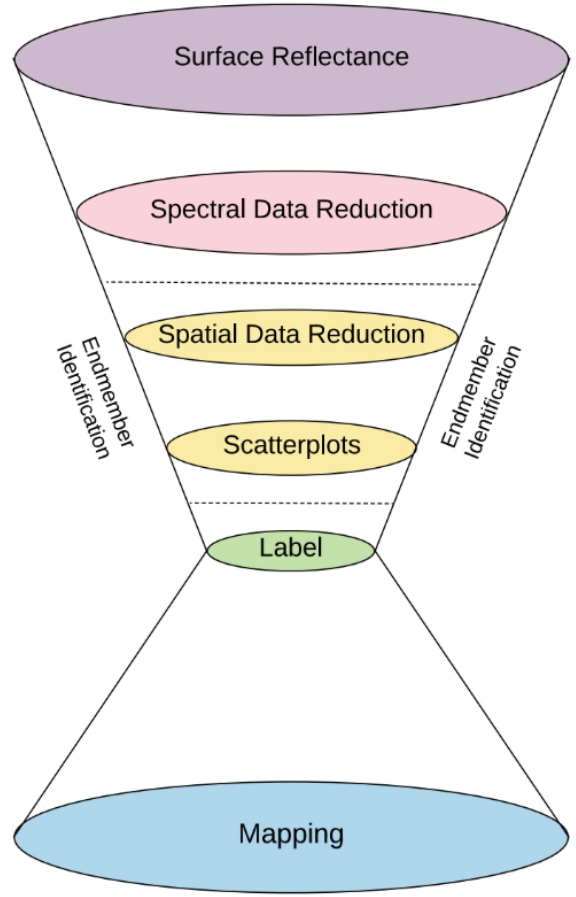


Landsat 9 sensor bands (blue) and asphalt spectrum (red).
https://landsat.usgs.gov/spectral-characteristics-viewer

From Spectra to Abundances

The next three slides explain MNF, endmembers, and unmixing. Surface reflectance is a prerequisite from Chapter 11.

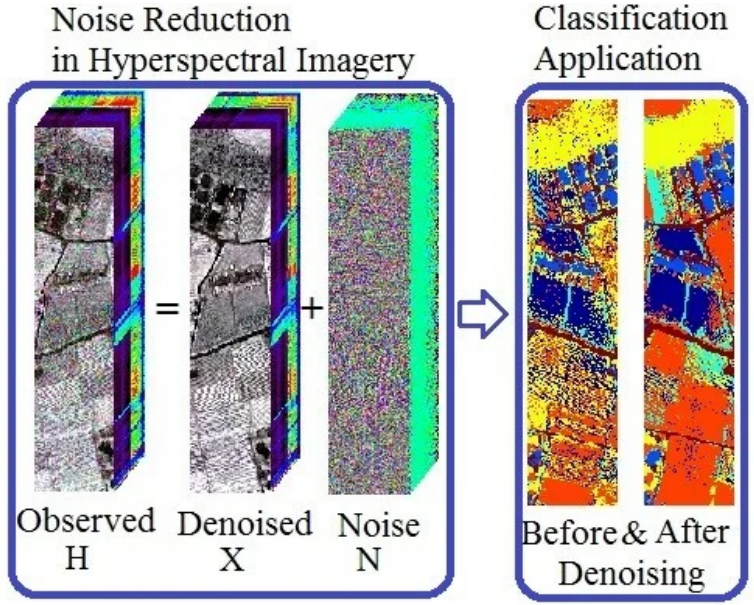
1. Start with atmospherically corrected surface reflectance.
2. Reduce noise dimensions using retained MNF components.
3. Select or estimate representative material spectra (endmembers).
4. Fit material fractions and inspect reconstruction residuals.

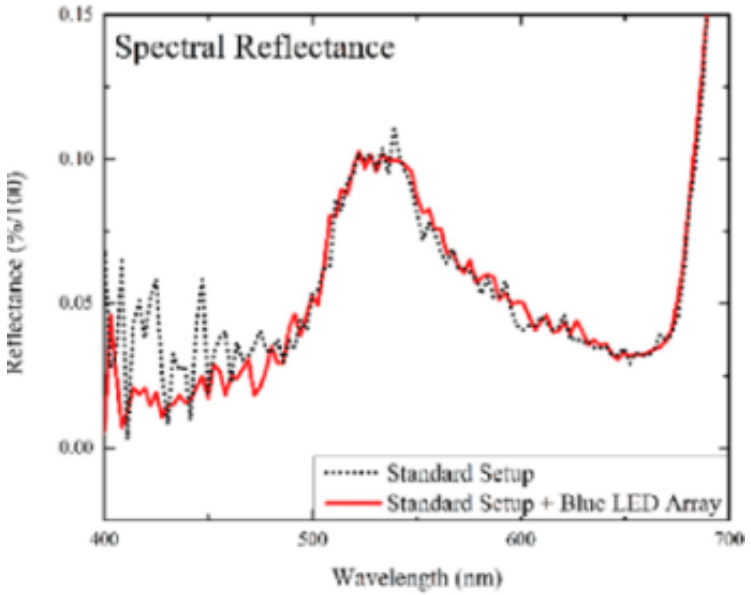


Maximum Noise Fraction (MNF)

- **Purpose**: Improve signal-to-noise by discarding noise-dominated components
- **Key steps**:
  i. Estimate noise statistics from sensor data or homogeneous regions.
  ii. Whiten estimated noise: transform its covariance to the identity. This does not remove noise.
  iii. Apply PCA: order transformed components by signal-to-noise.
- Analysts choose how many components to retain for material mapping.

MNF is a noise-aware extension of PCA.





MNF: Estimating and Suppressing Noise

Estimating the Noise Covariance Σ_n

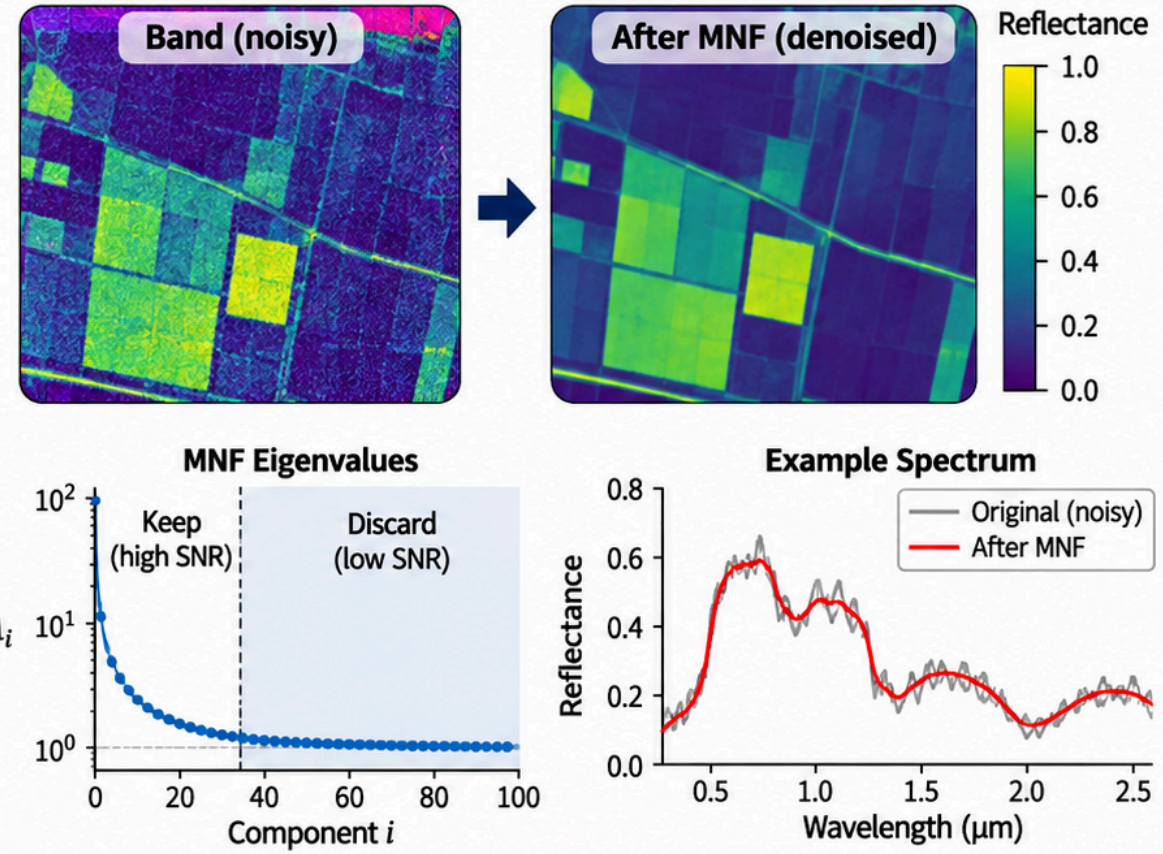
- Use differences between neighboring, similar pixels:

$$\boldsymbol{d} = \boldsymbol{x}_p - \boldsymbol{x}_q, \qquad \Sigma_n \approx \frac{1}{2} Cov(\boldsymbol{d})$$

- Assumes similar material ($s_p \approx s_q$) and independent noise.
- Alternatives: local regression residuals or sensor calibration.

MNF Denoising Procedure

1. **Whiten**: Transform with $W = \Sigma_n^{-1/2}$ so the noise covariance becomes the identity (I).
2. **PCA**: Apply PCA to the whitened data; components are ordered by signal-to-noise ($SNR_i = \lambda_i - 1$).
3. **Truncate & Reconstruct**: Keep the first k high-SNR components and inverse-transform to obtain denoised spectral bands.

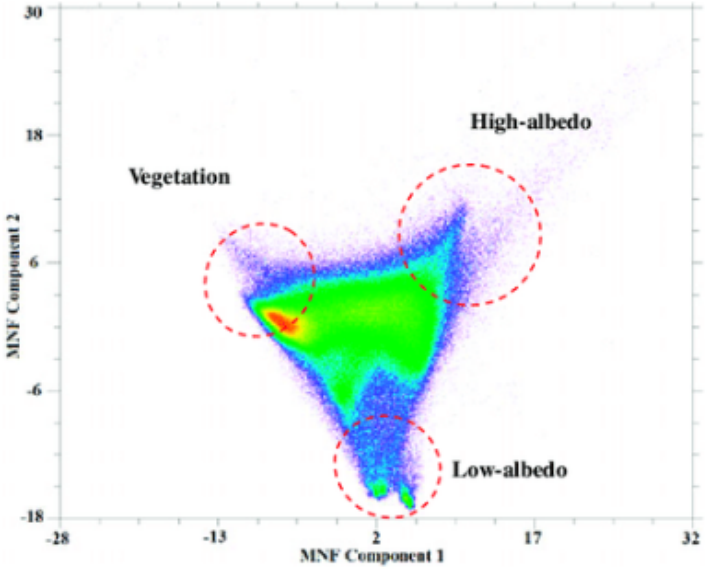


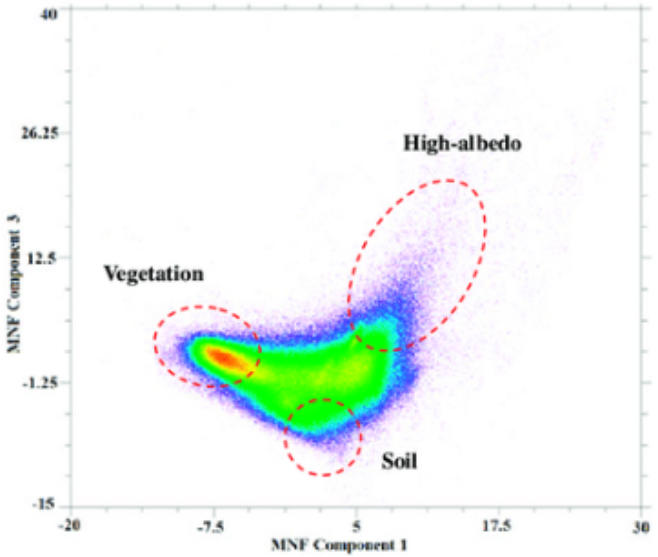
Takeaway: MNF uses an estimated noise covariance to whiten the data, orders components by SNR, and reduces noise by discarding low-SNR components.

AI-generated illustration

Pure Pixels and Spectral Endmembers

- A **scatter plot** projects each pixel into a feature space.
- **Endmembers**: spectra representing the materials used to explain scene pixels
- Pure pixels are useful candidates, but may not exist at the sensor's spatial resolution.
- Endmembers can also be estimated geometrically or statistically.
- Use libraries and field knowledge to interpret candidates; geometry alone does not name a material.





Linear Spectral (Un)mixing

- **Purpose**: Estimate material proportions in mixed pixels
- **Fraction vector $\mathbf{f}$**: proportions of each material
- Fully constrained fractions are **nonnegative** and **sum to 1**.

$$\mathbf{x} = \sum_{j=1}^{N} f_j \mathbf{m}_j = \mathbf{M}\mathbf{f}$$

$$f_j \geq 0, \qquad \sum_j f_j = 1$$

Columns of $\mathbf{M}$ are endmember spectra. We revisit this model with a residual term in Section 3.

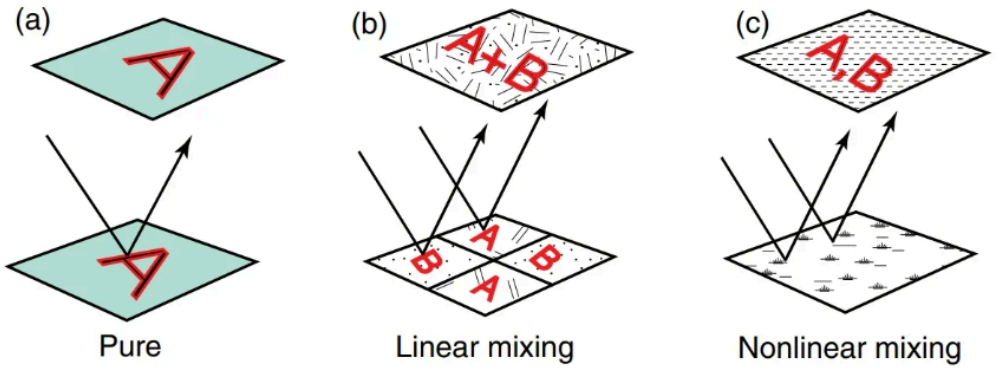


TABLE 14.1 Example of Mixture of Two Endmembers

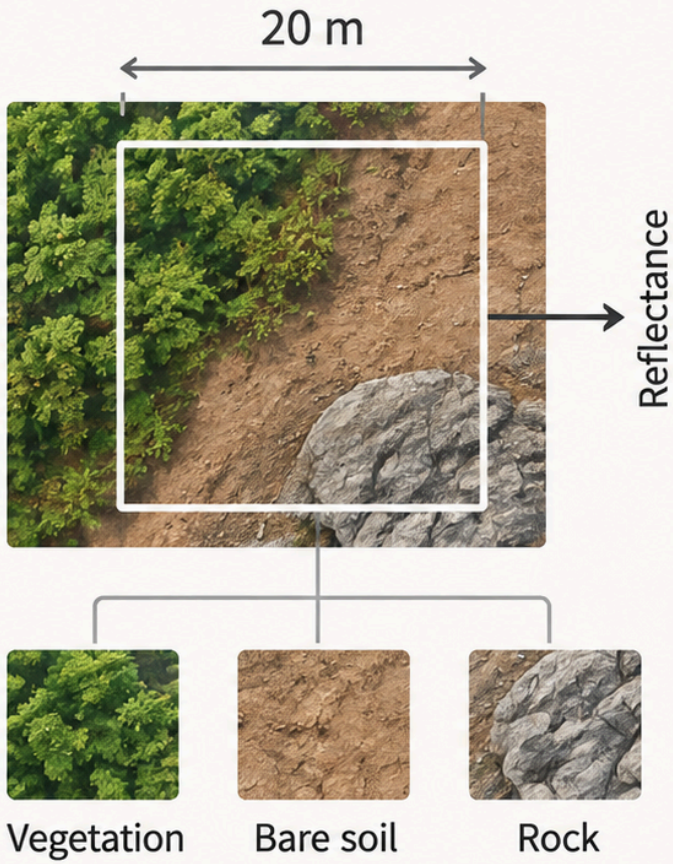
	Water	Trees	Riparian
Blue	1014.0	941.0	962.9
Red	832.0	799.0	808.9
Green	538.0	579.0	566.7
NIR	338.0	1846.0	1393.6

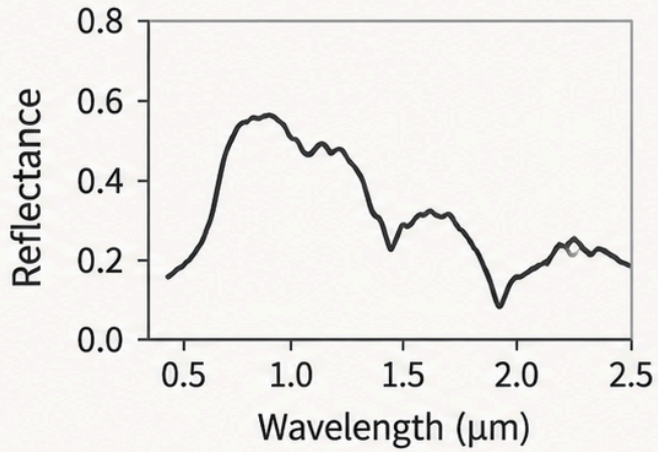
Section 3 | Chapter 14.8-14.10

Advanced Analysis

Endmembers, mixtures, and spectral matching

The Question Behind Unmixing





Single classification label

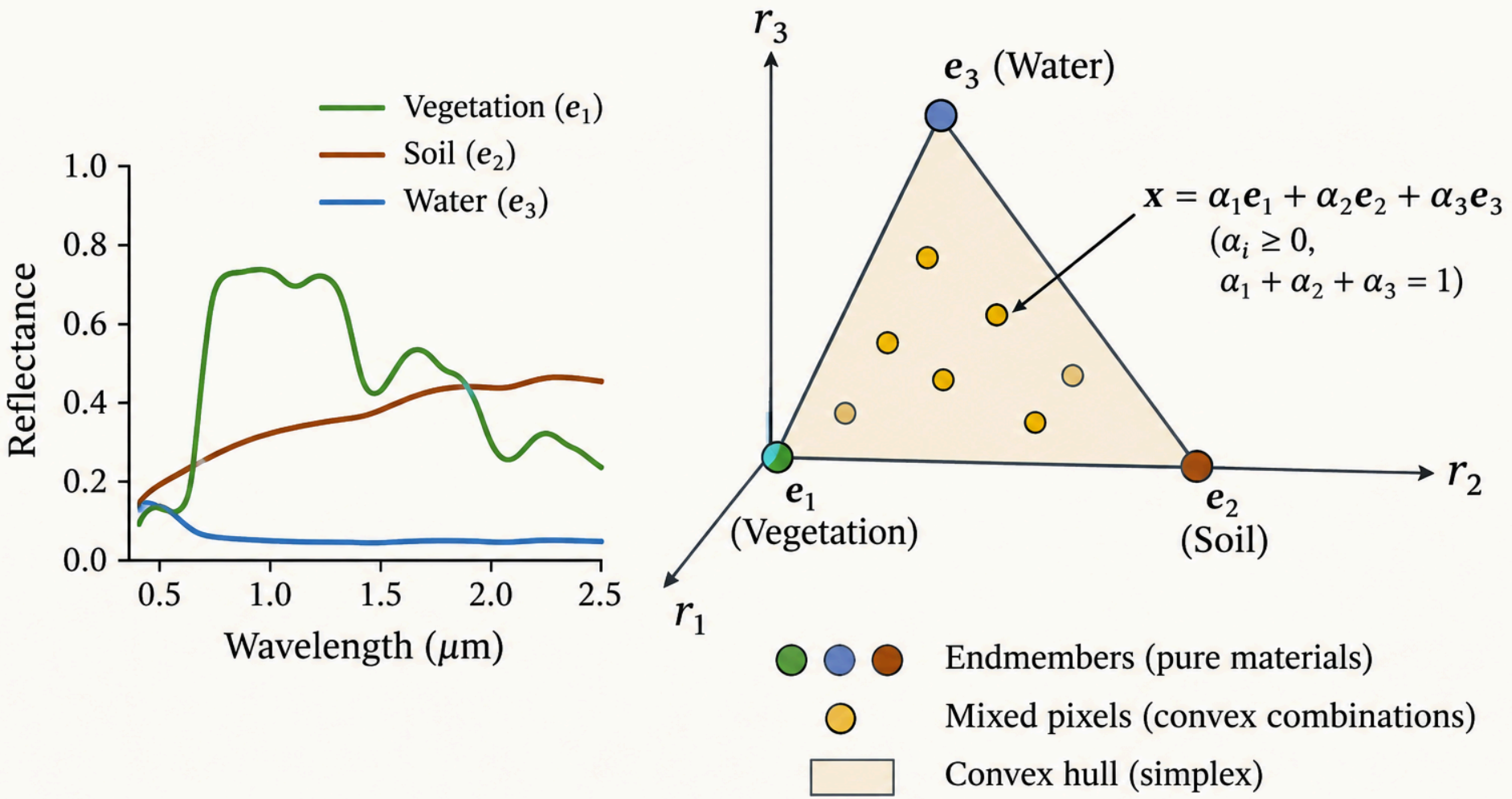
Vegetation

Material fractions (unmixing)

Vegetation	0.55
Bare soil	0.30
Rock	0.15

AI-generated illustration

Endmembers and Convex Geometry



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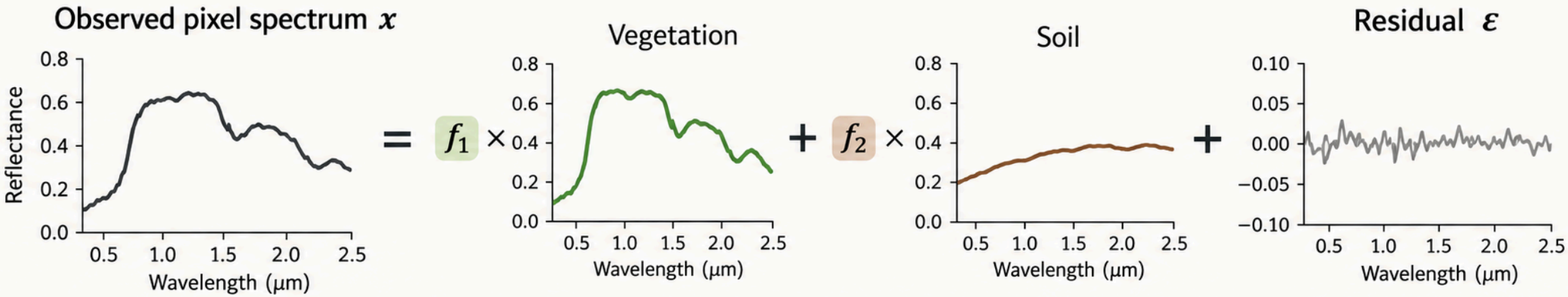
Linear Mixing Model: Including Residuals



$$\boldsymbol{x} = \boldsymbol{M}\boldsymbol{f} + \boldsymbol{\varepsilon}$$

$\boldsymbol{x}$: observed spectrum (L bands)

$\boldsymbol{M}$: endmember matrix $[\boldsymbol{m}_1 \quad \boldsymbol{m}_2]$

$\boldsymbol{f}$: abundance (fraction) vector $[f_1 \quad f_2]^T$

$\boldsymbol{\varepsilon}$: residual (modeling error, noise, variability)

$$f_1 \geq 0, \quad f_2 \geq 0, \quad f_1 + f_2 = 1$$

Vegetation endmember $\boldsymbol{m}_1$

Soil endmember $\boldsymbol{m}_2$

AI-generated illustration

A Small Mixture Example

Suppose a three-band pixel is a mixture of vegetation and soil:

$$\mathbf{m}_{veg} = [0.05, 0.10, 0.50]^T, \qquad \mathbf{m}_{soil} = [0.20, 0.25, 0.30]^T$$

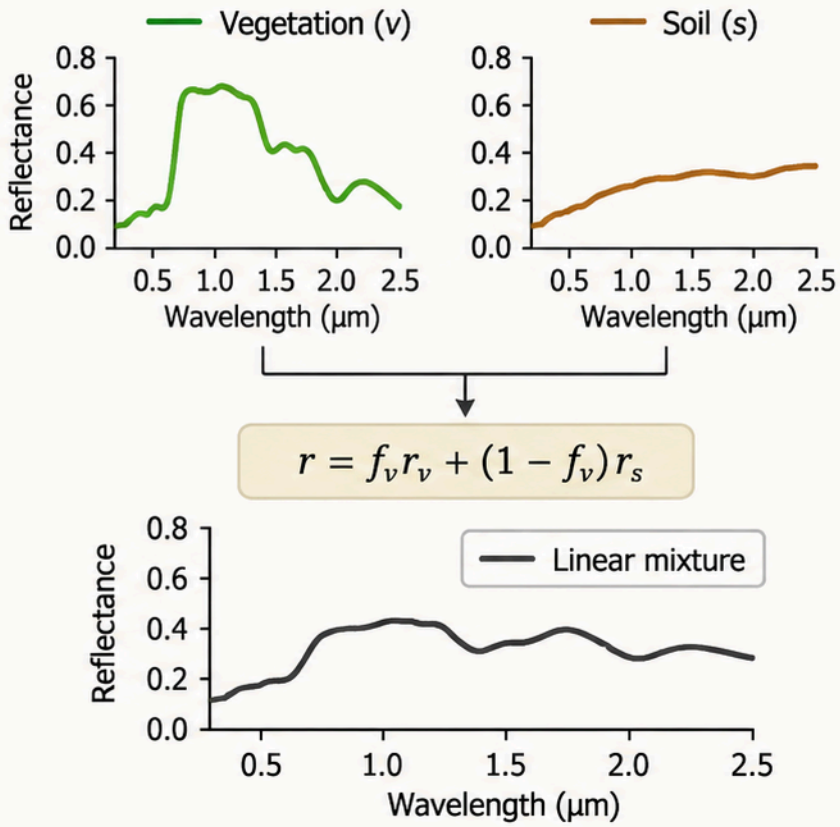
With fractions $\mathbf{f} = [0.7, 0.3]^T$:

$$\mathbf{x} = 0.7\mathbf{m}_{veg} + 0.3\mathbf{m}_{soil} = [0.095, 0.145, 0.440]^T$$

The estimate is only as meaningful as the endmembers and mixing model.

When Linear Mixing Breaks Down

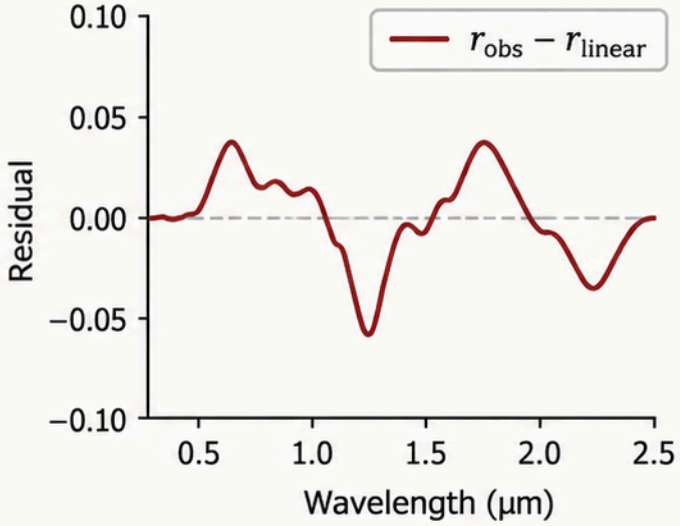
linear additive model



material interaction

multiple scattering
(within and between materials)

structured residual



AI-generated illustration

When Linear Mixing Breaks Down

- Linear mixing models side-by-side materials as an additive weighted sum, without material-to-material scattering interactions.
- Multiple scattering, intimate mineral mixtures, and layered canopies can violate that assumption.
- A structured residual is evidence to inspect, not a reason to add complexity automatically.
- Nonlinear models can represent interactions, but require additional assumptions or calibration data.

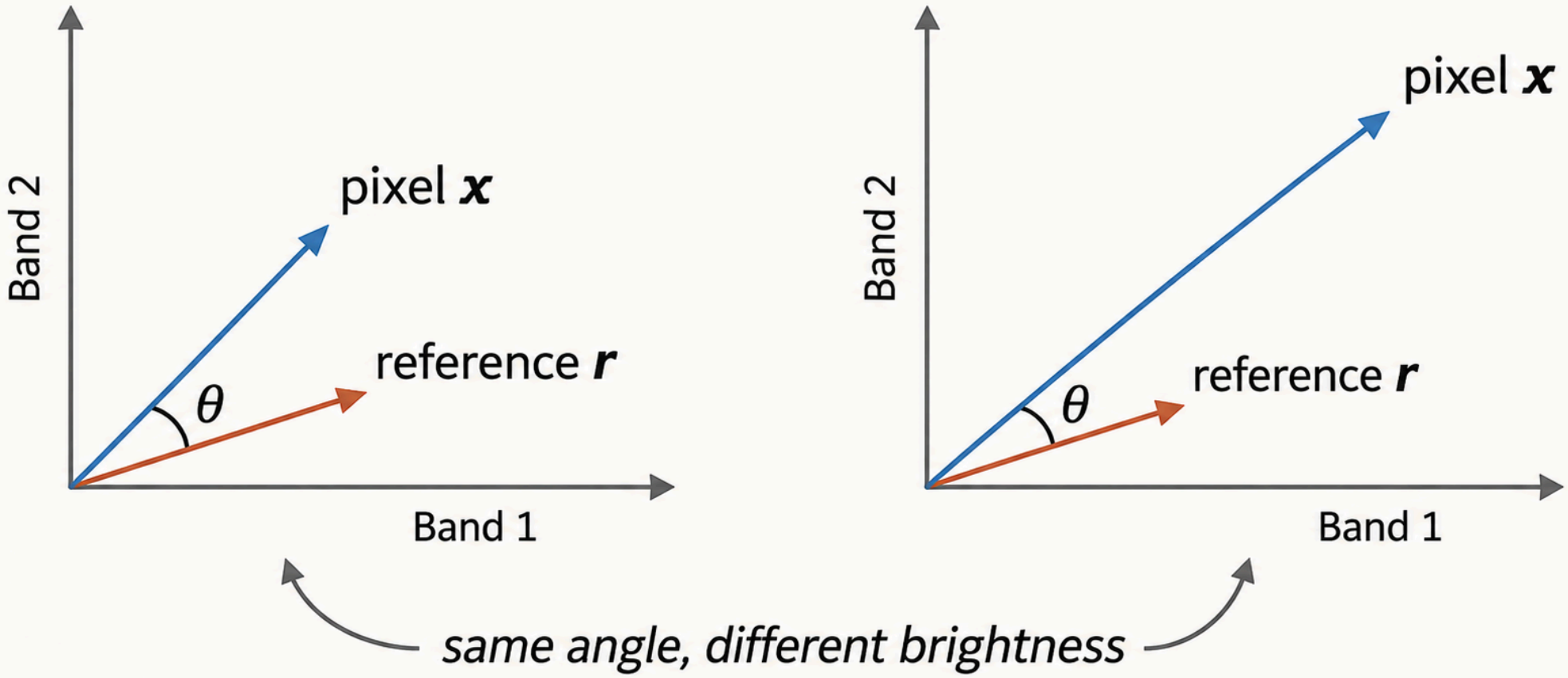
Spectral Angle Mapping

Compare a pixel spectrum with a reference spectrum by the angle between their vectors:

$$\theta = \cos^{-1}\left(\frac{\mathbf{x}^T\mathbf{r}}{\|\mathbf{x}\|_2\,\|\mathbf{r}\|_2}\right)$$

- Smaller angles indicate more similar spectral shape.
- Positive rescaling does not change the angle, so brightness differences matter less.
- SAM returns similarity to a reference; it does not estimate mixture fractions.

Spectral Angle Mapping



AI-generated illustration

Match, Classify, or Unmix?

Goal	Useful starting point	Main limitation
Match known material spectra	Spectral angle or library matching	Reference and scene conditions may differ
Assign one label per pixel	Supervised classification	Hides within-pixel mixtures
Estimate material proportions	Constrained unmixing	Depends on endmembers and mixing assumptions

Use the question and available reference data to choose the method.

Review: Putting the Analysis Together

1. Confirm wavelength calibration, bad-band handling, and reflectance units.
2. Reduce noise and inspect retained MNF components.
3. Select or estimate candidate endmembers; record their provenance.
4. Fit fractions or compare reference spectra.
5. Inspect residuals and spatial patterns before interpreting the map.

Multiscale Features and Learned Models

- Wavelets describe local spectral detail at different wavelength scales: broad continuum shape versus narrow absorption structure.
- The transform alone does not identify a material or correct the atmosphere; discarding coefficients also discards information.
- A supervised model such as an SVM can learn nonlinear decision boundaries from labeled spectra.
- A nonlinear classifier is not automatically a physical unmixing model; validate its predictions on independent samples.

Notebook: Fractions Versus Spectral Shape

Run the spectral mixture demo in Colab

View the notebook as slides

Recover known fractions, then dim a reference spectrum. Which result changes: its brightness, its spectral angle, or its material fractions?

Check Your Interpretation

A pixel has a small spectral angle to a vegetation reference, but its estimated vegetation fraction is low.

- Are these results necessarily contradictory?
- Which assumptions differ between angle matching and unmixing?
- What evidence would you inspect before naming the material?